# DNA Evolution via Directed Selection Inherits Paper 1's Modify Right

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## Attribution

*This work formalizes an inheritance edge between Paper 2 (Li, April 2026) and Paper 1 (Li, April 2026) of the Coordination Knowledge Substrate (CKS) theory series. It does not introduce new axioms. Its sole contribution is to establish, in precise form, that Paper 2's directed selection mechanism — the named governance process for deliberate DNA improvement — is the entity-DNA-scoped, improvement-oriented exercise of Paper 1's modify right, and that every structural property of the modify right travels into directed selection unchanged.*

## Abstract

Paper 1 of the Coordination Knowledge Substrate (CKS) series commits to the modify right as a structural property of human governance: humans can change any substrate content — including orchestration rules and behavior substrates — at any time, without requiring justification from or permission from any system component. Paper 2 formalizes directed selection as the named mechanism by which governance deliberately improves entity DNA specifications. This note establishes the inheritance edge C1.18: directed selection inherits the modify right. Every property of the modify right — human authority over what changes, no-justification-required, full attribution of the change, and at-any-time availability — is preserved without modification in directed selection. What Paper 2 adds is a named mechanism with its own governance record type, a version-history structure that accumulates successive directed selection events at entity scope, entity-level lineage chain participation where each improvement becomes part of the entity's governance history, a horizontal evolution variant where a successful improvement propagates to peer entities, and an improvement orientation that scopes directed selection specifically to governance-defined enhancements rather than arbitrary modifications. The note also positions directed selection as the governance-initiated counterpart to mutation (the subject of C1.17), completing Paper 2's two primary evolution mechanism types: environmental/external-managed and governance-initiated/internal-initiated. The prior-art significance is that named improvement mechanisms for entity specifications, DNA version

histories, lineage chain accumulation of modification events, and peer-propagation of improvements are all derivable from Paper 1's modify right and Paper 2's extension of it, foreclosing adversarial novelty claims in this space.

## 1. The inheritance edge

Paper 1's Claim 3 (A0.03) commits to the modify right as a structural human governance property. The commit is crisp: humans can change any substrate content — including orchestration rules and behavior substrates — at any time, without requiring justification or permission from any system component. The right is structural (always architecturally available, not contingent on workflow approval), immediate (exercisable at the time of the governance decision, not only at scheduled review windows), and authority-based (a governance decision is sufficient authorization; no system component can gate or deny it). Paper 1 also commits that modifications are attributed: who changed what, when, and under what authority is recorded.

Paper 2 formalizes directed selection as the named mechanism by which governance improves entity DNA specifications. Directed selection is a human-authorized governance event: governance identifies a specific improvement to a cell's, aspect's, or Self's DNA, authors the change, and records the change in the entity's DNA version history and lineage chain. Directed selection is governed through what Paper 2 describes as "the standard authority architecture from Paper 1" — meaning the modify right applies directly, with governance decision as sufficient authorization and attribution carried in the event record.

The inheritance relationship is therefore direct: directed selection is the modify right applied at entity-DNA scope with improvement orientation. It is not a new authority or a new class of permission. It is the modify right exercised at a specific architectural target (entity DNA) for a specific purpose (governance-defined improvement), with Paper 2 adding naming, versioning structure, and lineage chain participation around the exercise.

## 2. The modify right properties that travel unchanged

Four structural properties of Paper 1's modify right travel into directed selection without modification.

**Human authority over what changes.** No directed selection event occurs without governance authorization. There is no automatic DNA change mechanism in the CKS architecture: every change to entity DNA is human-initiated, human-authorized, and human-recorded. The LLM does not direct-select; it may participate as a drafting or analysis resource under human direction, but the authorization decision is governance's alone. This directly inherits the modify right's human-authority property.

**No-justification-required.** Governance can direct-select any DNA change it judges to be an improvement without requiring system approval, LLM agreement, or procedural sign-off from other architectural components. The governance decision is self-authorizing within the CKS authority architecture. This directly inherits the modify right's no-justification-required property. The architecture does not require governance to defend its improvement judgment to the system before the change takes effect.

**Attribution.** Every directed selection event carries full provenance metadata: who authorized the change, what content changed, when the event occurred, and at what scope (cell, aspect, or Self) the change applied. This attribution is not an audit add-on but a structural commitment — the event record is part of the entity's version history and lineage chain. The attribution property is directly inherited from Paper 1's commitment that modifications are attributed. What Paper 2 adds is the chain structure (§3 below), but the attribution metadata at each event node is a Paper 1 inheritance.

**At-any-time availability.** Governance can exercise directed selection at any point during an entity's existence — not only at scheduled review windows, release cycles, or designated improvement phases. The temporal property of the modify right (exercisable when governance decides, not when a workflow permits) carries directly into directed selection. An improvement identified outside any scheduled window can be authorized and recorded immediately.

These four properties together constitute the modify right's identity. Because all four travel into directed selection, directed selection is the modify right at entity-DNA scope, not an approximation of it.

## 3. What Paper 2 adds

Paper 2's contribution is not new authority — the authority was already in Paper 1's modify right. Paper 2's contribution is naming, structure, and scope-specific architecture around the exercise of that authority at entity-DNA scope. Five additions are distinct.

**Named mechanism.** Paper 1 established the modify right as a general governance property applicable to any substrate content. Paper 2 names the deliberate application of the modify right to entity DNA improvement as a specific mechanism — directed selection — with its own governance record type. Naming is architecturally significant: it makes the mechanism citable, testable, and governable as a distinct event class. Prior to Paper 2, the modify right was always available at DNA scope, but there was no named mechanism distinguishing deliberate improvement events from other substrate modifications. Directed selection is that name.

**DNA version history.** Directed selection events accumulate into a version history for the entity's DNA. Each change is versioned: the state of the DNA before the change, the change content, and the governance event record form a versioned entry. Successive directed selection events build a traversable history of the entity's DNA evolution under governance. This version-history structure is new in Paper 2. Paper 1 committed to attribution of individual modifications; it did not commit to an entity-level chain structure accumulating all modification events into a traversable history. The version history is built on Paper 1's attribution commitment (each entry carries the provenance metadata Paper 1 requires) but adds the chain structure that makes the history traversable and auditable as a sequence.

**Lineage chain participation.** Directed selection events are recorded in the entity's lineage chain — the entity-scope governance record that accumulates all events in the entity's history under governance. Each improvement becomes part of the entity's permanent governance record. This is distinct from the version history: the lineage chain records the entity's full governance history across all event types; directed selection events are a category within that chain. Paper 1 had attributed modifications at the substrate level; it did not have an entity-level lineage chain where each modification event participated as a named event type. Lineage chain participation is Paper 2's addition.

**Horizontal evolution variant.** Directed selection has a peer-propagation variant: when a successful improvement is identified for one entity, governance may apply the same directed selection to peer entities within the same aspect. This horizontal propagation — one improvement spreading across multiple entities at the same structural level — is new in Paper 2. Paper 1's modify right supported making the same modification to multiple entities, but there was no named horizontal evolution pattern for improvement propagation across peers. The horizontal variant is significant for defensive publication because it formalizes the governed peer-propagation shape explicitly.

**Improvement orientation.** Directed selection is oriented toward improvement: governance-defined enhancement of entity behavior. It is not just any modification to entity DNA but specifically a governed enhancement. This improvement orientation is new relative to Paper 1's modify right, which covered any change to substrate content — including corrections, reversions, structural reorganizations, and deletions. Directed selection's improvement orientation scopes the mechanism to a specific governance purpose, distinguishing it from other exercises of the modify right at DNA scope that do not have improvement as their purpose.

## 4. Contrast with C1.17: the two primary evolution mechanism types

Note C1.17 formalized the inheritance edge from Paper 2's mutation mechanism to Paper 1's tool-agnosticism commitment. Mutation in Paper 2 is environmental: it arises through external capability changes — LLM upgrades, substrate-platform infrastructure changes — that governance does not initiate but must manage. The LLM serves as the host layer for instinct capabilities, and changes to that layer arrive as undirected mutation events that governance responds to through verification substrates, routing rules, and retention or retirement decisions.

Directed selection, by contrast, is governance-initiated. Governance does not respond to a change that has already occurred; it decides that a change should occur, authors the change, and records it. The initiative is internal: governance identifies an improvement, authorizes it, and effects it through the modify right.

Together, mutation and directed selection cover the two primary evolution mechanism types Paper 2 establishes for entity DNA. Mutation is environmental/external-managed: capability changes arrive from outside, governance integrates them. Directed selection is governance-initiated/internal-initiated: governance decides an improvement is needed and makes it. Every DNA change in the CKS architecture is either a response to an environmental change (mutation) or a deliberate governance decision (directed selection). The two mechanisms are not alternatives — they operate in productive tension, with mutation providing exploratory capability the directed process could not have planned and directed selection providing stability and trajectory mutation alone could not deliver. The C1.17 and C1.18 inheritance edges together establish that both mechanism types are fully grounded in Paper 1 commitments: mutation in tool-agnosticism (the LLM-as-host-layer commitment), directed selection in the modify right (the human governance authority commitment).

The productive tension between mutation and directed selection is itself architecturally significant: it is the first point where CKS explicitly diverges from biology's evolved constraints. Biology has only undirected mutation; CKS has both undirected mutation and directed selection, held in governed productive tension. That Paper 1's commitments provide the full prior-art foundation for both mechanisms is what the C1.17–C1.18 pair establishes.

## 5. Prior-art significance

This inheritance edge forecloses three adversarial novelty claims.

**Claim (a): Named improvement mechanisms for entity specifications are novel relative to Paper 1's modify right.** This claim cannot succeed. Directed selection is the modify right applied at entity-DNA scope with improvement orientation. Every authority property of the modify right is present in directed selection unchanged. The naming and improvement orientation are additions Paper 2 makes, but the mechanism's authority structure is Paper 1's. A named improvement mechanism for entity specifications was architecturally available — and structurally committed — from Paper 1's modify right forward.

**Claim (b): DNA version histories and lineage chain accumulation of modification events are novel.** This claim cannot succeed. DNA version histories are built on Paper 1's attribution commitment: each version-history entry carries the provenance metadata (who changed what, when) that Paper 1's modify right already committed to. The chain structure is new in Paper 2, but it is constructed from attribution metadata Paper 1 already required. A party claiming novelty for version-history structures on attributed modification records would need to establish that Paper 1's attribution commitment did not already provide the necessary prior-art foundation, which it does.

**Claim (c): Peer-propagation of improvements (horizontal evolution) is novel relative to Paper 1's governed modification.** This claim cannot succeed. Paper 1's modify right was available at any scope and to any substrate content. Applying the same governance decision to peer entities was available under Paper 1's modify right from the paper's publication. Paper 2's horizontal evolution variant names and formalizes the peer-propagation shape, but the authority to make that shape available was already present in Paper 1.

## 6. Operational test

For any system claiming to implement directed selection as a CKS-coherent mechanism, the following three questions must all have affirmative answers.

First: can an observer verify that every directed selection event was human-authorized? This means: is there a governance record attached to each directed selection event identifying the human authority that authorized the change, such that no DNA change attributable to directed selection lacks a corresponding human authorization record? A system in which DNA changes can occur at entity scope without a human authorization record does not implement directed selection; it implements some form of automated or undefined modification that lacks the modify right's human-authority property.

Second: can an observer verify that the directed selection event is attributed in the entity's lineage chain? This means: does the entity's lineage chain contain the directed selection event as a named entry, with provenance metadata (who authorized, what changed, when) sufficient to reconstruct the change's governance context? A system that records the resulting DNA state but not the governed modification event does not implement the lineage chain participation Paper 2 commits to; it implements versioned storage without governance attribution, which is a weaker commitment than directed selection requires.

Third: can an observer verify that directed selection was available at the time of the change rather than only at a scheduled window? This means: is there evidence in the event record that the change was effected when governance decided to effect it, without requiring scheduling, approval-window gating, or workflow permission? A system that gates DNA modifications to scheduled review cycles satisfies a different governance pattern; it does not satisfy the at-any-time availability the modify right requires and that directed selection inherits.

A system that answers all three questions affirmatively for every directed selection event in its governance record implements the C1.18 inheritance commitment in full.

## 7. Conclusion

Directed selection inherits Paper 1's modify right because directed selection is the modify right applied at entity-DNA scope with improvement orientation. Human authority over what changes, no-justification-required, full attribution of each change, and at-any-time availability all travel from the modify right into directed selection unchanged. Paper 2's additions — naming, DNA version history, lineage chain participation, horizontal evolution variant, improvement orientation — build structure around the exercise of the modify right at entity-DNA scope. They do not create new authority; they use the existing authority to good effect at a new scope and for a specific governance purpose.

Positioned alongside C1.17 (mutation ⊃ tool-agnosticism), directed selection ⊃ human-governed authority completes the account of Paper 2's two primary evolution mechanism types. Both mechanism types are grounded in Paper 1 commitments: mutation in the substrate's tool-agnostic foundation, directed selection in the modify right's authority architecture. The productive tension between them — and the architectural result of holding both simultaneously under unified human governance — is a Paper 2 contribution built entirely on Paper 1 foundations.

## Source papers

*Li, W. (2026). Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems. April 2026. ORCID: 0009-0004-8065-3235.*

*Li, W. (2026). The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance. April 2026. ORCID: 0009-0004-8065-3235.*

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